linux电池管理

power-profile-daemon

`powerprofilesctl set` 查看当前选择的计划
`powerprofilesctl set <plan>` 选择plan电池计划预设

tlp

`sudo systemctl mask systemd-rfkill.service;sudo systemctl mask systemd-rfkill.socket;sudo systemctl enable tlp;systemctl start tlp`

`sudo tlp start` 启动auto预设

`sudo tlp usb` 启动usb自动挂载

tlp

Purpose

Apply TLP's settings and change mode of operation.

Start or restart TLP

Start TLP and apply power saving profile for the actual power source:

```
sudo tlp start
```

Note

Also use this command to apply changes after editing the configuration.

Battery Mode

Apply battery profile and enter manual mode:

```
sudo tlp bat
```

Hint: manual mode means that changes to the power source will be ignored until the next reboot or **tlp start** is issued to resume automatic mode.

AC Mode

Apply AC profile and enter manual mode:

```
sudo tlp ac
```

Hint: manual mode means that changes to the power source will be ignored until the next reboot or **tlp start** is issued to resume automatic mode.

USB Autosuspend

Apply autosuspend mode for all attached USB devices except those excluded by default or via configuration:

```
sudo tlp usb
```

Optical Drive

Power off optical drive in MediaBay/Ultrabay:

```
sudo tlp bayoff
```

Hints:

- Re-power the drive by releasing and reinserting the drive slot/Ultrabay eject lever; on newer models push the media eject button
- Devices other than optical drives – in particular hard disk drives – are not affected by this command

Battery Care

See also

- The level of battery care support as well as battery names and allowed charge threshold values depend on laptop vendor or brand, Linux kernel version and TLP version - consult Battery Care Vendor Specifics for details
- For further advice please visit the FAQ: Battery Care

Change battery charge thresholds temporarily

```
sudo tlp setcharge [<START_CHARGE_THRESH> <STOP_CHARGE_THRESH>] [BAT0|BAT1|BAT<x>|CMB0|CMB1]
```

Changes the charge thresholds for the battery to the given values.

Configured thresholds will be restored at the next boot or by using **tlp setcharge** again but without the threshold arguments.

Example:

```
sudo tlp setcharge 70 90 BAT0
```

Applies thresholds of 70/90% to battery *BAT0*.

Note

tlp setcharge changes the charge thresholds only temporarily. To make the change permanent, you must activate or change the related settings in the config file. Refer to Battery Care.

Charge battery to full capacity

```
sudo tlp fullcharge [BAT0|BAT1|BAT<x>|CMB0|CMB1]
```

This is done by applying vendor presets to the charge thresholds temporarily. Configured thresholds will be restored at the next boot or by using **tlp setcharge** without the threshold arguments.

Hint: after setting the thresholds the command terminates; it does not wait for the charge to complete.

Example:

```
sudo tlp fullcharge BAT1
```

Charges battery *BAT1* to full capacity.

Charge battery to the stop charge threshold once

ThinkPads only

```
sudo tlp chargeonce [BAT0|BAT1]
```

This is done by temporarily lifting the start charge threshold. The configured start charge threshold will be restored at the next boot or by using **tlp setcharge** without the threshold arguments.

Hint: after setting he thresholds the command terminates; it does not wait for the charge to complete.

Force a complete discharge of the battery while on AC power

ThinkPads only

```
sudo tlp discharge [BAT0|BAT1]
```

BAT0 selects the main/internal battery, *BAT1* the auxiliary/Ultrabay battery for discharge. The command continously shows remaining capacity and estimated discharge time. Discharging may be stopped at any time with `Control-C`.

Hints:

- The command needs the charger plugged in
- As soon as the battery is empty, charging begins and the command terminates; it does not wait for the charge to complete
- Normal use of the ThinkPad is possible during the discharge process; it does *not* suddenly switch off when the battery is empty
- ThinkPads with two batteries: the battery controller can only handle one battery at a time; while discharging one battery with this command the other battery can neither be charged nor discharged
- When encountering problems, see the FAQ: Battery Care

Perform a battery recalibration while on AC power

ThinkPads only

```
sudo tlp recalibrate [BAT0|BAT1]
```